

Oracle-Relative Resolution of Decision Partitions

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A decision task partitions hidden oracle instances into answer classes, but this kinematic partition alone does not determine query complexity. We give an exact one-query criterion in terms of the Gram matrix generated by the oracle family. For Deutsch’s task it yields a response-spectrum theorem: a controlled unitary response resolves the constant–balanced partition in one query if and only if the response has eigenvalue -1 . Under the standard Boolean point oracle, parity instead requires $\lceil N/2 \rceil$ quantum queries. We then enumerate all balanced binary partitions for three and four hidden bits, quotiented by oracle-preserving symmetries. All 35 three-bit tasks are certified exactly; 20 exhibit an exact advantage. For four bits, 395 of 6435 tasks are certified exactly and the remainder are separated into explicitly labeled numerical candidates. The results show that partitions specify what must be distinguished, whereas speedup depends on how efficiently the chosen oracle separates their cells.

OPERATIONAL CONTEXT AND KINEMATIC PARTITIONS

An oracle algorithm is often pictured informally as evaluating a function on many inputs in parallel. Operationally, however, it produces a single quantum state whose relative amplitudes can be recombined. What can be read out is determined by the distinguishability of the states associated with the possible oracle instances, not by a branchwise list of function values. This distinction is especially transparent in query problems, where the input is itself a black-box transformation.

It follows that a decision problem is not intrinsically one-query quantum or classically hard. Such statements necessarily refer to a task together with an oracle realization, an access convention, and an error criterion. Changing an oracle by a fixed basis transformation leaves quantum discrimination unchanged, whereas a different response spectrum, an instance-dependent phase, aggregated data, or an inaccessible environment can change the distinguishability produced by one call [1]. We first isolate this dependence for arbitrary finite unitary families and pointwise reversible responses, and then test the resulting picture by an exhaustive finite census.

Let Ω be a finite instance set and let $g : \Omega \rightarrow J$ specify the required answer. The task induces the kinematic partition

$$\Pi_g = \{C_j : j \in g(\Omega)\}, \quad C_j = g^{-1}(j). \quad (1)$$

This partition is useful bookkeeping, but it is not yet physics or complexity. In the query model the hidden instance labels a black-box transformation; the algorithm is not handed a state $|a\rangle$ on which the answer partition is a pre-existing observable. The operational question is whether the specified oracle and a fixed protocol can transfer the cell label to an accessible output.

More explicitly, let $\rho_a^{(T)}$ be the accessible state produced from instance a after T queries. Exact resolution means that there is one instance-independent measure-

ment $\{P_j\}$ satisfying

$$P_j P_\ell = \delta_{j\ell} P_j, \quad \sum_j P_j = I, \quad \text{Tr}(P_{g(a)} \rho_a^{(T)}) = 1. \quad (2)$$

Consequently, supports belonging to different cells are orthogonal; no orthogonality is required within a cell. The C_j themselves remain sets of classical instances. They are not subspaces, and the answer sectors $P_j \mathcal{H}_{\text{out}}$ arise only after an oracle and protocol have encoded them. The partition therefore specifies a required zero pattern in the output distinguishability matrix rather than a ready-made physical observable.

THE EXACT ONE-QUERY CRITERION

Consider a family of unitary oracles $\{U_a : a \in \Omega\}$. An arbitrary one-query protocol may prepare a pure state $|\Psi\rangle$ of the query system Q and a reference R , apply $U_a \otimes I_R$, and then use an instance-independent channel and measurement. Write

$$|\Psi_a\rangle = (U_a \otimes I_R) |\Psi\rangle, \quad \sigma = \text{Tr}_R |\Psi\rangle\langle\Psi|. \quad (3)$$

The output Gram matrix before the final processing is

$$G_{aa'} = \langle\Psi_a|\Psi_{a'}\rangle = \text{Tr}(\sigma U_a^\dagger U_{a'}). \quad (4)$$

Exact one-query criterion.— The partition Π_g is exactly resolvable with one use of $\{U_a\}$, allowing an arbitrary reference, if and only if there is a density operator σ such that

$$\text{Tr}(\sigma U_a^\dagger U_{a'}) = 0 \quad \text{whenever} \quad g(a) \neq g(a'). \quad (5)$$

Indeed, exact discrimination requires the spans of the output states from different cells to be orthogonal, which is equivalent to the indicated pairwise inner products vanishing. Conversely, orthogonal cell spans can be measured by their support projectors. Purification shows

that optimizing over σ already includes mixed probes and arbitrary ancillary systems. Hence Eq. (5), together with $\sigma \succeq 0$ and $\text{Tr } \sigma = 1$, is a semidefinite feasibility test, in the spirit of unitary-operation discrimination criteria [2, 3]. The Gram matrix need only be block diagonal according to Π_g ; states within one cell may overlap or coincide. Complete identification is the special case in which every cell is a singleton and G must be diagonal. For bounded error, orthogonality is replaced by the existence of a common POVM with the required success probability. For standard input-oracle models, multiquery state conversion and function evaluation are characterized by adversary methods [4, 5].

Write $Q_\epsilon(\Pi; \mathcal{U})$ and $R_\epsilon(\Pi; \mathcal{U}_{\text{cl}})$ for the optimal quantum and classical worst-case numbers of calls needed with worst-case error at most ϵ , under explicitly matched access conventions. Equation (5) characterizes $Q_0 = 1$; it does not by itself establish an advantage. Speedup additionally requires a gap between Q_ϵ and R_ϵ for the same task and oracle interface. Classically, a deterministic query protocol partitions Ω by its possible transcripts and succeeds exactly only when that transcript partition refines Π_g . A quantum protocol need not create such an intermediate classical transcript; it changes the overlaps among the hypothesis states and measures only after the required cells have separated. The relevant resource is this rate of separation, not the number of formal components in the probe superposition.

The criterion also distinguishes a change of representation from a change of access. If

$$V_a = AU_aB \quad (6)$$

for instance-independent unitaries A, B , then $V_a^\dagger V_{a'} = B^\dagger U_a^\dagger U_{a'} B$; as the probe varies, the attainable Gram matrices are unchanged. More generally, families that exactly simulate one another with $O(1)$ oracle calls have quantum query complexities equal up to constant factors; an approximate simulation must also charge its conversion error to ϵ . Preservation of a claimed quantum-classical gap additionally requires corresponding constant-query classical simulations. Exact one-query behavior can nevertheless change when the response representation is not of the form Eq. (6).

For example, Hadamard conjugation of the answer qubit turns Eq. (7) below into the full controlled-phase action $|x, z\rangle \mapsto (-1)^{z b_x} |x, z\rangle$, so the two forms are exactly equivalent. The restricted sign oracle $P_b |x\rangle = (-1)^{b_x} |x\rangle$ is only a subroutine: $P_{b \oplus 1} = -P_b$, and the complementary truth tables differ by a global phase. Supplying a coherent control for P_b restores the missing relative phase, but coherent control is additional black-box access rather than an automatic operation on an unknown unitary.

DEUTSCH'S PROBLEM AND THE RESPONSE SPECTRUM

Deutsch's problem illustrates both the block criterion and this dependence. Let $b = (b_0, b_1)$ be the truth table of a Boolean function and use

$$O_b |x, y\rangle = |x, y \oplus b_x\rangle. \quad (7)$$

With the answer qubit in $|-\rangle$, phase kickback gives, up to the common answer factor, the output states on the address register:

$$\begin{aligned} |\psi_{00}\rangle &= |+\rangle, & |\psi_{11}\rangle &= -|+\rangle, \\ |\psi_{01}\rangle &= |-\rangle, & |\psi_{10}\rangle &= -|-\rangle. \end{aligned} \quad (8)$$

Thus the constant and balanced cells occupy orthogonal rays, while the two instances inside either cell are not distinguished. This is precisely a block-diagonal, but not diagonal, Gram matrix and gives the familiar one versus two exact-query separation [6].

The phenomenon is controlled by the response spectrum. Let V be any unitary and consider the controlled-response family indexed by the same four Boolean truth tables,

$$U_b^{(V)} = \sum_{x=0}^1 |x\rangle\langle x| \otimes V^{b_x}. \quad (9)$$

Response-spectrum criterion.—The Deutsch constant-balanced partition is exactly one-query resolvable through $\{U_b^{(V)}\}$, with arbitrary probes and references, if and only if -1 is an eigenvalue of V .

For necessity, write the probe as $|0\rangle|\phi_0\rangle + |1\rangle|\phi_1\rangle$, with $|\phi_x\rangle$ on the response and reference systems. Let $\rho_x = \text{Tr}_R |\phi_x\rangle\langle\phi_x| \succeq 0$ be its unnormalized response operator and put

$$p_x = \text{Tr } \rho_x, \quad z_x = \text{Tr}(\rho_x V), \quad p_0 + p_1 = 1. \quad (10)$$

The four cross-cell orthogonality conditions include their complex conjugates and reduce to

$$z_0 = z_0^* = -p_1, \quad z_1 = z_1^* = -p_0. \quad (11)$$

Since $|z_x| \leq p_x$, these equations force $p_0 = p_1 = 1/2$ and $z_0 = z_1 = -1/2$. Equality in the unitary expectation bound is possible only when the support of each nonzero ρ_x lies in the -1 eigenspace of V . Conversely, a -1 eigenvector provides ordinary phase kickback and Eq. (8).

For the elementary cyclic response $X_d |y\rangle = |y + 1 \bmod d\rangle$, $d \geq 2$, computational-basis queries return the same Boolean point value for every d (by mapping $y \mapsto y + b_x \pmod{d}$ with $b_x \in \{0, 1\}$), but -1 occurs in the spectrum of X_d exactly when d is even. Therefore

$$Q_0^{(d)} = \begin{cases} 1, & d \text{ even}, \\ 2, & d \text{ odd}, \end{cases} \quad R_0^{(d)} = 2. \quad (12)$$

In particular, reversible qutrit addition removes Deutsch’s exact quantum advantage without changing its classical point values. This is a constant-factor effect, not an asymptotic separation, but it demonstrates that reversibility and blank-register behavior do not determine coherent query power.

GENERAL POINTWISE PARITY

The obstruction persists more strongly for general parity under the standard point oracle. Let $b \in \{0, 1\}^N$ and $O_b |i, y\rangle = |i, y \oplus b_i\rangle$. Every pure probe, including a reference system, can be written in the answer qubit’s X basis as

$$|\Psi\rangle = \sum_{i=1}^N |i\rangle (|+\rangle |\alpha_i\rangle + |-\rangle |\beta_i\rangle), \quad p_i = \|\beta_i\|^2. \quad (13)$$

The overlap between the outputs for 0^N and the odd string e_i is

$$\langle \Psi_{0^N} | \Psi_{e_i} \rangle = 1 - 2p_i. \quad (14)$$

Exact even-odd separation would require $p_i = 1/2$ for every i , while normalization gives $\sum_i p_i \leq 1$. Hence one query is possible only for $N \leq 2$, with Deutsch as the limiting case. More generally, the acceptance probability of a T -query algorithm is a polynomial of degree at most $2T$ in the input bits, whereas parity has exact degree N . Thus $T \geq N/2$, and applying the Deutsch construction to disjoint pairs gives the matching algorithm, so

$$Q_0(\text{Par}_N) = \lceil N/2 \rceil, \quad R_0(\text{Par}_N) = N \quad (15)$$

for point-value access [7, 8]. A binary partition can therefore remain query intensive. If instead one declares $\tilde{O}_b |y\rangle = |y \oplus \text{Par}(b)\rangle$ to be a single oracle call, parity becomes one-query resolvable classically as well as quantum mechanically. That does not defeat Eq. (15); it replaces point access by an oracle that already aggregates the desired answer.

BALANCED-PARTITION CENSUS

The preceding examples suggest that the partition is a task specification, while cost is controlled by its relation to the available queries. This can be tested exhaustively for small point oracles. Let the hidden instance be $b \in \{0, 1\}^N$, accessed through Eq. (7) with the address range enlarged to N , and consider every binary answer map having 2^{N-1} instances in each cell. Exchanging the answer labels gives the same unlabeled partition, so their number is

$$M_N = \frac{1}{2} \binom{2^N}{2^{N-1}}. \quad (16)$$

Coordinate permutations, independent complements of coordinates, and answer exchange preserve the point-oracle query cost. They reduce the $M_3 = 35$ partitions to six orbits and the $M_4 = 6435$ partitions to 58. Orbit sizes are restored when reporting the counts below.

For each partition we computed the optimal deterministic decision-tree depth D and sought matching bounds on the exact quantum query complexity Q_E . One-query feasibility reduces, for the Boolean response, to rational Gram constraints and was solved over the rationals. Multiquery feasibility was formulated with the Gram-matrix semidefinite program of Barnum, Saks, and Szegedy [9]. Constructive upper bounds were retained as adaptive trees whose nodes read either one bit or the parity of two bits; the latter is one Deutsch query. Such trees are useful certificates, but they are not a general characterization of exact quantum algorithms [10].

For $N = 3$ all lower and upper certificates meet exactly:

(D, Q_E)	(1, 1)	(2, 1)	(2, 2)	(3, 2)
partitions	3	3	12	17.

(17)

Thus 20 of the 35 balanced tasks admit an exact advantage: three by a factor of two and seventeen by a factor of $3/2$. The certificates also display the conditional structure hidden by the bare partition. For example, three-bit majority is decided by first obtaining $b_1 \oplus b_2$; if it is zero one reads b_1 , and if it is one one reads b_3 . Likewise, $F = b_1 \oplus b_2 b_3$ is decided by first reading b_2 , followed by b_1 when $b_2 = 0$ and by the Deutsch parity $b_1 \oplus b_3$ when $b_2 = 1$. Both real multilinear polynomials have degree three, so the polynomial method excludes one query and proves these two-query constructions optimal.

For $N = 4$, exact rational lower certificates and explicit trees settle 395 partitions: 4, 6, 48, 248, and 89 have $(D, Q_E) = (1, 1), (2, 1), (2, 2), (3, 2)$, and $(4, 2)$, respectively. Of these, 343 have an exact advantage. For the remaining 6040 partitions, exact trees give $Q_E \leq 3$, while two SCS runs at requested tolerances 10^{-7} and 10^{-8} report the two-query SDP infeasible. The resulting assignments, 1536 at $(D, Q_E) = (3, 3)$ and 4504 at $(4, 3)$, are therefore *numerical candidates*, not exact lower bounds. If accepted, they give candidate totals of 4847 speedup and 1588 no-speedup partitions. Exact dual certificates are required before these totals can be stated as theorems.

Numerical SDP studies of all Boolean functions through four bits were already reported in Ref. [10]. The purpose of the present census is narrower: it reorganizes the balanced functions as unlabeled decision partitions, quotients only by symmetries that preserve point access, supplies simple exact certificates where available, and keeps numerical status explicit. The observed proportions are consequently finite-model facts, not an asymptotic speedup criterion.

BERNSTEIN–VAZIRANI AND INACCESSIBLE RECORDS

Bernstein–Vazirani supplies the opposite extreme. For $a, x \in \mathbb{F}_2^n$, let $f_a(x) = a \cdot x \pmod{2}$. One phase query prepares the character state

$$|\chi_a\rangle = 2^{-n/2} \sum_x (-1)^{a \cdot x} |x\rangle, \quad \langle \chi_a | \chi_{a'} \rangle = \delta_{aa'}. \quad (18)$$

The singleton partition is therefore resolved in one query; the final Hadamards merely map $|\chi_a\rangle$ to $|a\rangle$ [11]. This orthogonality can be degraded without changing any computational-basis value returned by the oracle.

Choose $S \subseteq \{1, \dots, n\}$, $|S| = k$, and let x_S denote the corresponding substring. Consider the unitary dilation

$$W_a^S |x, y, e\rangle = |x, y \oplus a \cdot x, e \oplus x_S\rangle. \quad (19)$$

Each call receives a fresh witness in $|0^k\rangle$, which is discarded afterward. The accessible oracle is consequently the channel

$$\Phi_a^S(\rho) = \text{Tr}_E[W_a^S(\rho \otimes |0^k\rangle\langle 0^k|)(W_a^S)^\dagger]. \quad (20)$$

Basis queries still return $a \cdot x$ exactly, but the witness records selected address information and removes coherences between sectors with different x_S , as in standard which-way interference [12]. With the usual Bernstein–Vazirani input and phase kickback, tracing out the witness leaves

$$\rho_a^S = 2^{-n} \sum_{x, x' : x_S = x'_S} (-1)^{a \cdot (x \oplus x')} |x\rangle\langle x'|. \quad (21)$$

Character orthogonality after the final Hadamards gives, up to a fixed reordering of coordinates,

$$H^{\otimes n} \rho_a^S H^{\otimes n} = |a_{\bar{S}}\rangle\langle a_{\bar{S}}| \otimes \frac{I_S}{2^k}, \quad (22)$$

$$\Pr(z|a) = 2^{-k} \mathbf{1}[z_{\bar{S}} = a_{\bar{S}}].$$

Thus this one-query readout exactly resolves

$$\Pi_S = \{C_u : u \in \mathbb{F}_2^{n-k}\}, \quad C_u = \{a : a_{\bar{S}} = u\}. \quad (23)$$

For $k < n$ and any worst-case error $\epsilon < 1/2$, this yields the oracle-relative comparison

$$Q_\epsilon^\Phi(\Pi_S) = 1, \quad R_\epsilon^{\text{val}}(\Pi_S) = n - k. \quad (24)$$

Here R_ϵ^{val} denotes randomized computational-basis value-query complexity. Zero queries cannot resolve the nontrivial partition Π_S , because the output would be independent of its cell label. The classical upper bound uses the basis queries outside S . For the lower bound, even impose the favorable promise $a_{\bar{S}} = 0$. Under the uniform distribution on $a_{\bar{S}}$, fewer than $n - k$ returned linear equations leave an affine solution space of positive

dimension, so even a decoder given the complete transcript succeeds with probability at most $1/2$. The same average bound holds after including the protocol’s internal randomness, and therefore some input has success at most $1/2$. Equation (24) therefore concerns the same surviving partition and the same computational-basis value interface. It does not claim that the inaccessible witness is free: if the witness were coherently available, the fixed coupling $|x, e\rangle \mapsto |x, e \oplus x_S\rangle$ could be undone without another oracle call, restoring the ordinary Bernstein–Vazirani output. The accessible oracle here is a channel, not the globally unitary dilation. Related distinctions between standard and garbage-bearing oracle access arise throughout quantum query complexity [13].

At $k = 0$, no address coordinate is recorded and Eq. (22) is the ordinary Bernstein–Vazirani output resolving the singleton partition, giving the n -versus-one identification gap. At $k = n$, the stated interference readout is uniform and Π_S has only one cell, although ordinary basis queries can still recover a . The model separates loss of the phase-interference route from loss of the classical function values themselves.

The loss also admits a continuous form. For each $i \in S$, let the two witness records have real overlap $\langle \epsilon_0^{(i)} | \epsilon_1^{(i)} \rangle = \lambda_i \in [0, 1]$, and use the product record $|\epsilon_{x_S}\rangle = \bigotimes_{i \in S} |\epsilon_{x_i}^{(i)}\rangle$. The map $|x, y, 0\rangle \mapsto |x, y \oplus a \cdot x\rangle |\epsilon_{x_S}\rangle$ is an isometry on the initialized subspace, and hence admits a unitary extension, because the address x is retained. The same calculation gives

$$\Pr_\lambda(z|a) = \mathbf{1}[z_{\bar{S}} = a_{\bar{S}}] \prod_{i \in S} \frac{1 + (-1)^{z_i \oplus a_i} \lambda_i}{2}. \quad (25)$$

The stated one-query readout recovers the full string with probability $\prod_{i \in S} (1 + \lambda_i)/2$. The endpoints $\lambda_i = 1$ and 0 represent, respectively, fully retained coherence and a perfectly distinguishable witness record; intermediate overlaps interpolate the visibility and success probability of the displayed Bernstein–Vazirani readout continuously. This probability characterizes that probe and Hadamard readout, not the optimal one-query strategy for the complete query–answer channel.

CONCLUSION

Equation (1) specifies which hidden instances may share an answer, but it neither supplies an input state $|a\rangle$ nor selects a physical output observable. The oracle and protocol determine how rapidly states from different cells separate. The response-spectrum theorem and the contrast between point and aggregate parity show that changing access changes this separation even when the abstract task is fixed. Conversely, the finite census holds the point oracle fixed while varying every balanced target. Its exact three-bit certificates reveal ad-

vantages built from oracle-accessible, conditionally composed refinements, while the four-bit calculation marks the boundary between proof and numerical evidence.

Shor’s and Grover’s algorithms are relevant to the partition picture, but at different levels. In period finding, fix the modular-exponentiation oracle U_a and partition its address values into the cosets $C_t(r) = \{x : x \equiv t \pmod{r}\}$ of the unknown period r . The oracle response is constant on each such cell. Coherent modular exponentiation and the quantum Fourier transform do not read a pre-existing period label; they create and then resolve this algebraic partition statistically, allowing r to be inferred [14]. The subsequent classical step maps an estimated period to factors. This is a structured partition–oracle pair, not an arbitrary balanced partition of the type enumerated in our finite census.

Grover gives the complementary example. Under the promise of either no mark or exactly one mark, let $\Omega = \{\emptyset\} \cup \{t : t \in [M]\}$. The decision partition has the two cells $\{\emptyset\}$ and $\{t : t \in [M]\}$, while the search task refines the latter into singleton cells. The marking oracle $O_t|x\rangle = (-1)^{[x=t]}|x\rangle$ supplies only a small separation per call; amplitude amplification accumulates it in $\Theta(\sqrt{M})$ bounded-error queries rather than the classical $\Theta(M)$, and this black-box scaling is optimal [15, 16]. The speedup therefore does not come from a finer partition. It comes from the geometry and symmetry of the oracle family, which let one coherent operation act on all candidate cells at once.

Partitions are therefore indispensable as specifications but insufficient as complexity criteria. Quantum advantage is not caused by partition coarseness or by simultaneous access to imagined branch values. It occurs when structure-matched interference separates the required answer cells in fewer calls than any classical transcript under the same interface. Establishing an asymptotic speedup requires a structured family of such task–oracle pairs; percentages from a finite census cannot substitute for that analysis.

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Data availability.—The source code, exact certificates, and generated tables underlying the finite census are available in the [public supplementary archive](#) and from the author upon reasonable request.

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